BDA400 Assignment 5

Technical Analysis using R - Development Phase

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Course: BDA400 - Data Science Tools and Techniques
Assignment: A05

Repository location expected from Assignment 2:
https://github.com/mahmuudbakkar-pixel/TechnicalAnalysis

The URL above came from the prior Assignment 2 package. This environment could not verify that the repository exists or upload Assignment 5 to it. The folder in the ZIP is ready to add as **TechnicalAnalysis/Assignment5**; verify the link after uploading.

Implementation overview

Nine separate R files implement the supplied function templates with base R. The scripts use no technical-analysis libraries. `run_tests.R` sources them in dependency order and checks ordinary and boundary cases. `analyze_portfolio.R` is an optional application to the actual historical CSV files from the preliminary stage.

Function / script	Method and result
sma(data, period) - sma.R	Average each complete window. The output has n - period + 1 observations.
ema(data, period) - ema.R	Start with the first observation, then update with multiplier 2/(period + 1).
macd(data, short_period, long_period, signal_period) - macd.R	Subtract long EMA from short EMA; smooth the difference for the signal line; subtract again for the histogram. Returns a three-element named list.
stdev(data) - stdev.R	Square deviations from the mean and divide by n before taking a square root; this follows the handout population formula.
linreg(regressionSource, regressionLength, regressionOffset) - linreg.R	Select an exact-length window ending at the requested offset; calculate least-squares slope, intercept and predicted values with position indexes.

Indicator implementations, continued

Function / script	Method and result
rsi(data, period) - rsi.R	Separate upward and downward price changes, seed average gain/loss from the first period differences, then apply Wilder smoothing. Leading values are NA. A flat series returns 50; only gains return 100.
stoch_rsi(data, period, k_period, d_period) - stoch_rsi.R	Within each complete rolling RSI window, normalize current RSI between local minimum and maximum (%K); %D is a moving average of complete %K values. Both lines align with input positions.
crossover(arr1, arr2) - crossover.R	Compare current and prior paired observations; return None, Up or Down, matching the pseudocode.
crossunder(arr1, arr2) - crossunder.R	Detect movement from at/above to below; return None at the first position and True or False strings afterward, matching the pseudocode.

Ambiguities resolved from the handout

The regression pseudocode start position creates one extra observation; the implementation uses end - regressionLength + 1. The crossover descriptions alternate between Boolean values and string labels; the scripts follow the explicit pseudocode labels. Stochastic RSI uses a rolling range and aligned output so its current value cannot depend on future RSI observations. A constant rolling RSI window is set to neutral 0.5 to avoid division by zero.

How to run and verify

Open the Assignment5 directory in RStudio or a terminal, then run **Rscript run_tests.R**. The supplied **test_results.txt** records that all nine indicators passed normal and edge case checks with R 4.3.3. Source `ema.R` before `macd.R`, and `rsi.R` before `stoch_rsi.R`. All test inputs are synthetic examples and are not represented as market prices.

Actual market-data application

Use the AAPL, MSFT, GOOGL, AMZN and TSLA symbols from Assignment 2. Place genuine historical OHLCV CSV exports in `market_data/`, one file per symbol, with Date, Open, High, Low, Close and Volume fields. Running **Rscript analyze_portfolio.R** then creates a data frame for each available stock, filters the most recent 365 calendar days, calculates indicators, and writes illustrative signals and a stock filter. No actual market CSVs accompanied this assignment, so no real-stock output is claimed in this submission.

Source: CDI College, BDA400 Assignment 5 - Technical Analysis using R, Development Phase, pp. 2-20 (provided handout). Portfolio symbols and expected repository name: Douha Suliman Assignment 2 package.